

SUBJECT: Artificial Intelligence

ASSIGNMENT # 01 DUE DATE

NAME: Kumard Supraja

NIST ROLL # 202361046 SECTION 'A'

Writing space begins here. Do not leave this Page blank.

- 1) Describe the attributes of intelligent agent. Explain diff^t types of environments in which agents have to operate

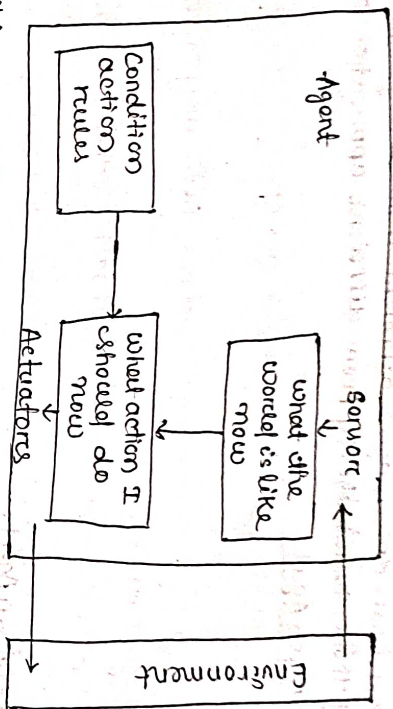
Attributes of Intelligent Agent:

- * Autonomous Control: The agent operates independently without external control, making decisions based on its programming and collected information.
- * Ability to perceive its Environment: It gather data from its surrounding using sensor or other output.
- * Ability to adapt to changes: The agent can adjust its actions or strategies when the environment changes, ensuring it continues to work towards its goal.
- * Pursue its Goals: It has specific objective and takes action that move it closer to achieving these goals.
- * Persistence Over time: The agent continuously operates over a prolonged period, consistently working towards its objectives instead of acting just once.
- * Striving for the best outcome: Even when there is uncertainty the agent uses its knowledge to make the best decisions possible for best outcome.

Types of Environments for Agents:

- 1) Fully observable vs partially observable:

- Fully observable: Agent can sense the complete state.
Eg: chess board
- Partially observable: Only partial information is available. Eg: Driving.



Functionality:

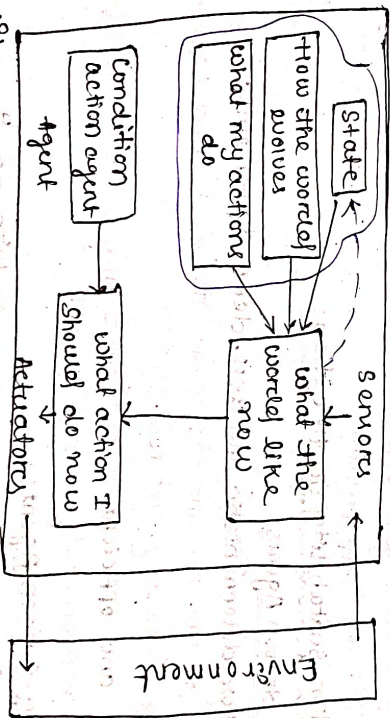
- Respond to environment changes in real-time.
- Operate without complex reasoning or planning.
- The reactive agent have specific actions that are predefined and associated with particular conditions or events.

Eg: Autonomous vacuum cleaners, Basic robotex arms (operate using direct commands and sensory input for tasks like picking and placing objects).

Deliberative Agents:

Structure:

- more complex architecture.
- Incorporate knowledge representation and reasoning capabilities.
- Maintain an internal state and model of the environment.



Functionality:

- Capable of planning and decision making over a longer time horizon.
- utilize reasoning to evaluate different actions based on goals and potential outcome.
- Can adapt strategies based on past experience and predictions.

Eg: chess programs, self-driving car

- Essential for real-world decision making systems that require flexibility, adaptability and optimization of multiple criteria.

5) Perform a critical analysis of Iterative Deepening and Iterative Broadening search techniques with their applicability.

1. Iterative Deepening Search:

- Iterative deepening search is a DFS algorithm that repeatedly runs a depth-limited DFS, increasing the depth limit with each iteration until a solution is found.
- Essentially, IDS combines the benefits of DFS and BFS

Characteristics:

- Space complexity: $O(b \cdot d)$ where, b = branching factor
 d = depth
- Time complexity: $O(b^d)$

- Optimality: If the cost of all steps is uniform, IDS finds the shallowest solution and is optimal.

- Completeness: IDS is complete if the search space is finite and all goal nodes are reachable.

Applicability:

- Suitable for problems where the depth of the solution is not known in advance.
- Works well in tree search problems where memory limitations are a concern but the solutions depth is unknown.

2. Iterative Broadening Search:

- Iterative broadening search is derived from BFS in which the search expands only a limited number of child nodes at each depth level, gradually increasing the breadth of the search.
- Unlike BFS or IDS, which control the depth, IBS control, IBS controls the breadth of the search.

Characteristics:

- Space complexity: IBS is typically more space-efficient than BFS.

- Time complexity: $O(b^d)$

- Optimality: IBS does not guarantee optimality as limiting the breadth could mean skipping optimal solutions that exist in early iterations.
- Completeness:

IBS can be incomplete because by limiting the breadth, some goal nodes could be missed in earlier iterations, leading to the failure to find a solution.

Applicability:

- Useful in situations with massive branching factors, where it is impractical to explore all children at once.
- Practical for problems where exhaustive search is impractical, such as certain heuristic searches, scheduling problems or AI decision-making systems in environments with huge state spaces but limited computational resources.

6) What is the role of heuristics in informed search techniques?

Explain the A* algorithm with an example.

- Heuristics play a crucial role in informed search techniques, as they provide additional information about the problem to guide the search process more efficiently.
- A heuristic is essentially a function that estimates the cost or distance from a given node to the goal.
- Informed search algorithms use this estimate to prioritize which paths or nodes to explore first, leading to faster and more efficient solutions, especially in larger or complex search spaces.

A* algorithm:

- The A* algorithm is a popular informed search technique that combines the strengths of both uniform cost search and greedy best-first search.
- It uses both the actual cost to reach a node ($g(n)$) and an estimated cost to reach the goal from that node ($h(n)$) to determine which node to explore next.

$$f(n) = g(n) + h(n)$$

where,

$g(n)$ = The actual cost from first node to node n .

$h(n)$ = The estimated cost from node n to the goal.

7) State the 3 missionaries and 3 cannibals trying to cross a river as a constraint satisfaction problem. Provide a possible solution to the problem.

- 3 missionaries & 3 cannibals were on the side of the river. Au want to cross river.

- On each side of river missionaries can't be less than the cannibals.
- Only one boat is available. It carry only two people at a time.
- And the boat don't cross the river itself with no people on board.

Left

0 0 0 X X X

missionaries cannibals

< 1, 3, 3 >

Return

0 0 0 X <

< 1, 3, 1 >

0 0 0 X X

< 1, 3, 2 >

Return

0 0 0 <

< 1, 3, 0 >

0 0 0 X

< 1, 3, 1 >

Return

0 X <

< 1, 1, 1 >

0 0 X X

< 1, 2, 2 >

Return

X X <

< 1, 0, 2 >

X X X

< 1, 0, 3 >

X <

< 1, 0, 1 >

X X

< 1, 0, 2 >

Right

< 2, 0, 0 > → Initial state

X X

< 2, 0, 2 >

X

< 2, 0, 1 >

X X X

< 2, 0, 3 >

X X

< 2, 0, 2 >

X X 0 0

< 2, 2, 2 >

X 0

< 2, 1, 1 >

X 0 0 0

< 2, 3, 1 >

0 0 0

< 2, 3, 0 >

X X 0 0 0

< 2, 3, 2 >

X 0 0 0

< 2, 3, 3 >

X 0 0 0

→ Intermediate State

$\langle 1, 0, 0 \rangle$

000'x'x'

$\langle 2, 3, 3 \rangle$

↓
Goal state

Q) What is the significance of Adversarial Search? How does alpha-beta pruning help?

- Adversarial search is a type of search used in environments where multiple agents are competing against each other, and the outcome for one agent depends on the decisions made by the others.

- This type of search is typically used in games like Chess, Tic-tac-toe.

Significance Of Adversarial:

- Adversarial search is critical in competitive environments where two or more agents have conflicting goals and compete against each other.
- This form of search models decision-making in situations where each player's move affects the other player's outcome, which makes it essential in games, artificial intelligence (AI), and strategic decision-making scenarios.

Alpha-Beta Pruning:

- Alpha-beta pruning is an optimization technique used to improve the efficiency of the minimax algorithm by "pruning" parts of the game tree that don't need to be explored because they cannot influence the final decision.

Working of Alpha-Beta Pruning:

Alpha-beta pruning eliminates branches that will not be considered in the final decision, allowing the search to focus only on relevant parts of the tree.

° Alpha (α): Represents the best value that the maximizing player can guarantee so far!

° Beta (β): Represents the best value that the minimizing player can guarantee so far.

If the current node cannot improve the score for the current player, further exploration of that branch is unnecessary and can be "pruned".

Q) What is the role of predicate logic in representing knowledge for AI systems? How does it help in inferring new knowledge?

Role of Predicate Logic in Representing Knowledge for AI systems

- Predicate logic is a powerful formalism used to represent knowledge in AI systems.
- It extends propositional logic by allowing the representation of more complex structures, such as objects, their properties, and the relationships between them.
- Predicate logic plays a crucial role in AI because it enables:

1. Expressive Representation:

- Predicate logic allows AI systems to represent more detailed and structured knowledge about the world. Instead of dealing with simple true/false values, as in propositional logic, predicate logic can express knowledge about objects, their attributes and their interactions.

2. Use of Quantifiers:

- Predicate logic introduces quantifiers, such as universal quantifiers and existential quantifiers which allow AI systems to represent generalized knowledge or specific instances.

3. Representation of Relations:

- In predicate logic, you can represent relationships between different objects. This is critical for knowledge representation in AI, where the focus is often on how entities interact or relate to one another.

How Predicate Logic helps in Inferring new knowledge

- Predicate logic not only represents knowledge but also allows AI systems to infer new knowledge using logical inference rules.
- Inference is the process by which AI systems deduce new information based on existing facts and logical rules.

1. Unification: Unification is the process of matching two predicate logic statements by finding a substitution of variables that makes the two statements identical. It is critical for reasoning systems to match rules with specific facts.

2. Resolution: Resolution is a fundamental rule of inference in predicate logic. It is used in automated reasoning and theorem proving to derive conclusions by eliminating contradiction. Resolution works by combining clauses that contain complementary literals to produce new conclusions.

3. Backward and Forward Chaining:

- o Forward chaining: starts with known facts and applies inference rules to derive new facts until a goal is reached. Used in data-driven applications like expert systems.

- o Backward chaining: starts with the goal and works backward to determine which facts or rules lead to that goal. Used in goal-directed reasoning like query answering.

10) Explain the use of semantic net in knowledge representation. Develop a property inheritance algorithm to retrieve the value of an attribute for an object belonging to a particular class.

Use of semantic networks in knowledge representation

- A semantic network is a graph-based structure used to represent knowledge in AI systems. It consists of nodes representing the relationships between these concepts and edges representing relationships between these concepts.
- Semantic network provides a visual and structured way to model knowledge, emphasizing the relationships and hierarchy between objects and their attributes.
- They are often used in areas such as natural language processing, expert systems and knowledge management.